EXERCISE 4

Question 1

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
       ProjectID, ProjectName, Budget, Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Complete
2	Alice	Smith	HR	60000	102	Employee	50000	ongoing
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial a	90000	ongoing
5	Emma	Wilson	Sales	65000	105	Marketing	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending

Question 2

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
       ProjectID, ProjectName, Budget, Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
2	Alice	Smith	HR	60000	102	employee	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial	90000	ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Marketing	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending
1	John	Doe	IT	70000	101	AI Development	100000	Complete

Question 3

```
SELECT ProjectID, ProjectName, Budget, Status,
       EmployeeID, FirstName, LastName, Department, Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
ON B.EmployeeID = A.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
103	Cybersecurity	75000	Pending	1	John	Doe	IT	70000
102	Employee training	50000	ongoing	2	Alice	Smith	HR	60000
104	Financial audits	90000	ongoing	3	Bob	Johnson	Finance	75000
NULL	NULL	NULL	NULL	4	David	Brown	IT	72000
105	Marketexpansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk management	80000	Pending	6	Michael	Clark	Finance	80000
101	AI development	100000	Completed	1	John	Doe	IT	70000

Question 4

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
       ProjectID, ProjectName, Budget, Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
3	Bob	Johnson	Finance	75000	104	Financial	90000	ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending
2	Alice	Smith	HR	60000	102	Employee training	50000	ongoing
5	Emma	Wilson	Sales	65000	105	Marketexp	65000	Completed
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
1	John	Doe	IT	70000	101	AI development	100 000	Completed

Question 7

SELECT EmployeeID, firstname, lastname, Department, Salary
FROM Employees
WHERE ~~Salary~~ salary > 70000;

EmployeeID	firstname	lastname	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Question 8

SELECT EmployeeID, firstname, lastname, Department, Salary
FROM Employees
WHERE Department ~~IT~~ = 'IT' OR Department = 'Finance';

EmployeeID	firstname	lastname	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Question 9

SELECT ProjectID, ProjectName, Budget, status
FROM Projects
WHERE NOT status = 'Completed';

ProjectID	Projectname	Budget	Status
102	Employee training	50000	ongoing
103	Cybersecurity	75000	Pending
104	Financial and	90000	Ongoing
106	Risk management	80000	Pending

Question 10

```
SELECT ProjectID, Projectname, Budget, Status
FROM Projects
WHERE status IN ('ongoing', 'Pending') AND Budget
>70000;
```

ProjectID	Projectname	Budget	Status
103	Cybersecurity	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk management	80000	Pending

Question 11

```
SELECT EmployeeID, firstname, lastname, Department, Salary, City
FROM employees
WHERE city IN ('New-york', 'Toronto')
ORDER By Salary DESC;
```

EmployeeID	firstname	lastname	Department	Salary	City
6	Michael	Clark	Finance	80000	New-york
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New-york

Question 12

```
SELECT EmployeeID, firstname, lastname, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;
```

EmployeeID	firstname	lastname	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

Question 13

```
SELECT Department,
        SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

Question 14

```
SELECT City,
        AVG(Salary) AS Average Salary
FROM employees
GROUP BY City
```

HAVING AverageSalary > 65000;

City	AverageSalary
New-York	70000
Toronto	75000
London	72000

Question 15

```
SELECT Department,
        COUNT(EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY department
HAVING EmployeeCount > 1;
```

Department	EmployeeCount
IT	2
Finance	2

Question 16

```
SELECT Status,
        COUNT(ProjectID) AS ProjectCount
FROM projects
GROUP BY Status
HAVING ProjectCount >= 2;
```

Status	ProjectCount
Completed	2
ongoing	2
Pending	2

Question 17

```
SELECT EmployeeID, Firstname, Lastname,
        SUM(Budget) AS TotalProjectBudget
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID, Firstname, Lastname
HAVING TotalProjectBudget >150 000;
```

EmployeeID	Firstname	Lastname	TotalProjectBudget
1	John	Doe	175000